



Introduction to MPEG-G

The Genomic Data Compression Standard



Background

MPEG-G is an international standard for genomics data management, covering compression, security and privacy as well as overall data interchange and data management.

- **Uniform Data Language:** Standardization means our genomic data speaks the same language across the board, removing the constant need for translation. This lets machine learning models leverage larger, richer datasets, enhancing their accuracy and clinical relevance.
- **Faster, Smarter Storage and Models:** With data ready to go, AI models can more swiftly predict, diagnose, and tailor treatments, improving efficiency and outcomes in healthcare settings.
- **Open Collaboration:** Easier and more secure data sharing under standardized protocols boosts worldwide collaboration among scientists and clinicians, accelerating research and the spread of innovative solutions.





The Genomic Data Compression Standard

- MPEG-G is an ISO/IEC standard for genomic data compression and transport.
- It enables efficient storage, selective access, and streaming of genomic data.
- Supports sequencing reads (FASTQ), alignments (BAM), and variants (VCF).
- Promotes interoperability and scalability for clinical and research genomics.
- Enables privacy controls, selective encryption, and metadata tagging.

Why MPEG-G Matters

- 5–10x compression over BAM/FASTQ formats reduces storage costs.
- Enables fast, targeted decompression for specific genomic regions.
- Facilitates data sharing across institutions with standardized containers.
- Includes security features like selective encryption and user-based access.
- Ideal for clinical genomics, population studies, and AI model training.



Genie: Open Reference Software for MPEG-G

- Genie is an open-source encoder/decoder for MPEG-G.
- Developed by Leibniz University Hannover and other contributors.
- Enables compression of FASTQ, BAM, and VCF files into MPEG-G.
- Supports partial file access, region-based decompression, and metadata indexing.
- Available on GitHub: github.com/mpeg-g/genie



More about MPEG-Genomics

- › Müntefering, F., Adhisantoso, Y.G., Chandak, S. *et al.* Genie: the first open-source ISO/IEC encoder for genomic data. *Nature Commun Biol* **7**, 553 (2024).
<https://doi.org/10.1038/s42003-024-06249-8>
- › [Genie: the first open-source ISO/IEC encoder for genomic data](#)
- › [MPEG-Genomics Users on LinkedIn](#)
- › [MPEG Meetings](#)
- › [WG 8 – MPEG Genomic Coding](#)

